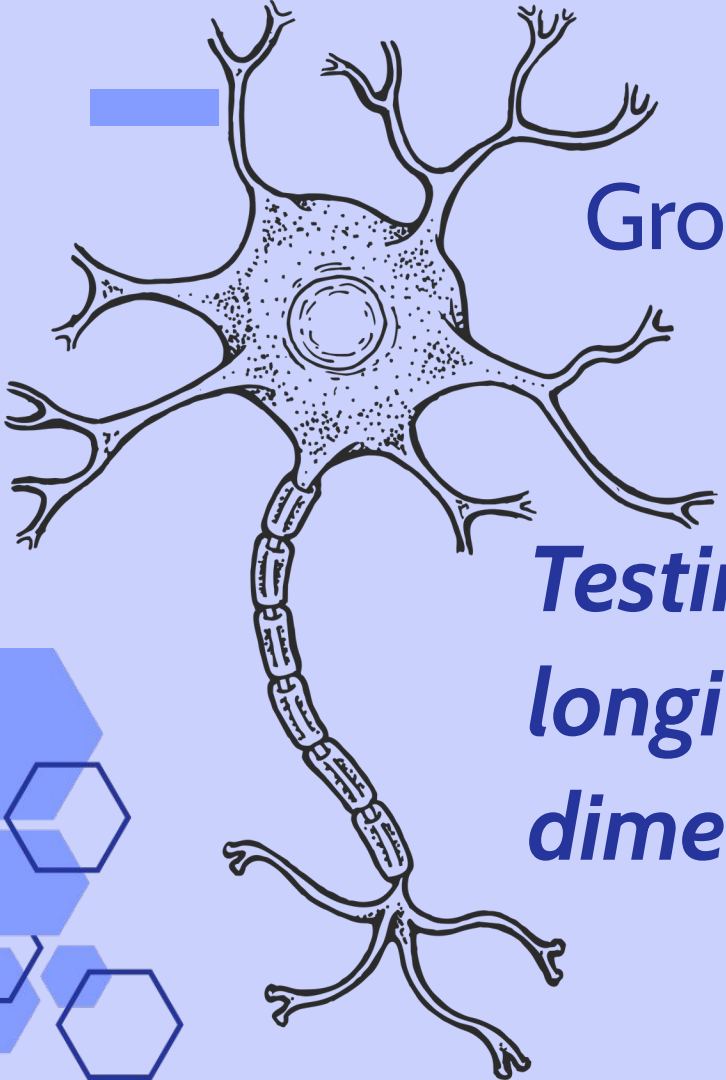
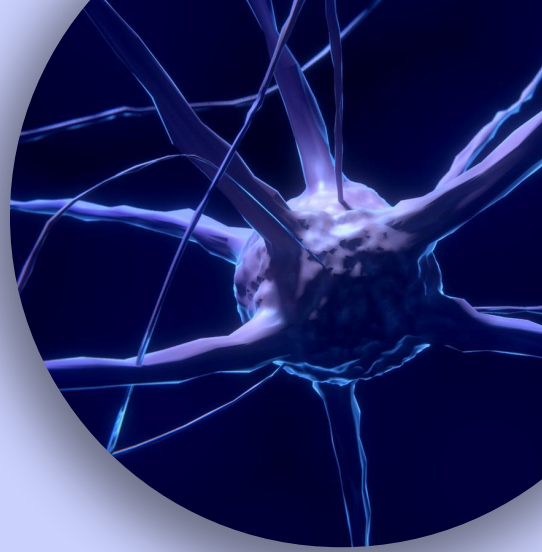


Group:NeuroNexus



Testing one versus two longitudinal PD progression dimensions in PPMI

Tanya Sharma, Shibasis Patnaik, Tomoya Kubota

Can early Parkinson's disease progression be represented by one latent stage, or are two progression dimensions needed?

We will test whether repeated cognitive, dopaminergic, and structural measurements in early Parkinson's disease can be adequately represented by a single latent longitudinal disease stage, or whether the data are better explained by two correlated progression dimensions within the same individual.

One-stage model

- One common PD progression dimension

Two-stage model

- Two correlated progression dimensions

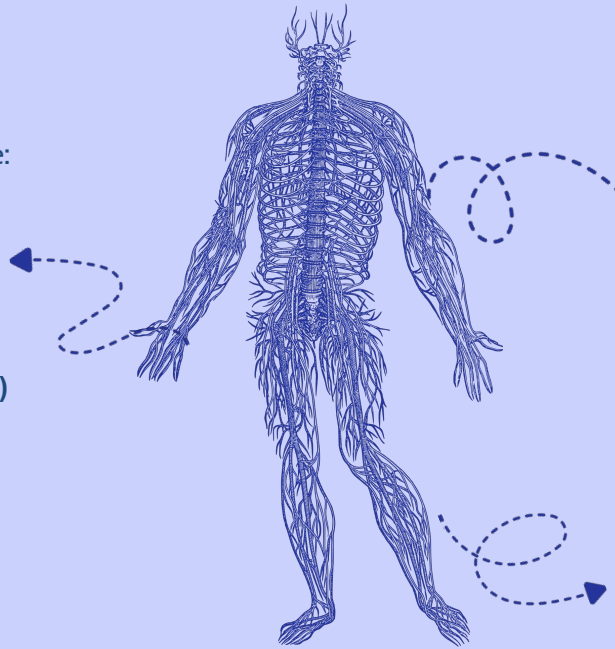
Primary question: Does the two-stage model predict later, unseen measurements better than the one-stage model?

Why move beyond the PPMI Cognitive Summary Score?

What is the CSS?

CSS combines six robustly normed cognitive measures into a single score:

HVLT-R Immediate
HVLT-R Delayed
Judgment of Line Orientation (JLO)
Letter–Number Sequencing (LNS)
Symbol Digit Modalities Test (SDMT)
Semantic Fluency



Why use it?

The CSS is an established PPMI cognitive benchmark and showed useful **cross-sectional discrimination between newly diagnosed PD participants and robust healthy controls**.

What remains unresolved?

The six tests do not necessarily change in the same way over time, and combining them into one score does not establish that **longitudinal cognition or PD progression is one-dimensional**.

Our question: When cognition is examined alongside DaTScan and structural MRI, is progression better represented by one shared dimension or two partially separable dimensions?

Weintraub et al., 2025; Turner et al., 2023; González et al., 2025

Preliminary PPMI data audit

N= 1,112

Participants with repeated complete **six-test cognition**



N= 865

Participants with repeated **DaTScan**

DaTScan: Screening · M12 · M24 · M48

N= 137

Participants with repeated processed **grey-matter volume**

Processed GMV: Baseline · M24 · M48

N= 82

Participants with longitudinal information across cognition, DaTScan, and structural MRI

Because the modalities follow different schedules, the analysis will use actual scan and assessment dates rather than assuming that nominal study visits are synchronized.

Data source: Parkinson's Progression Markers Initiative (PPMI), ppmi-info.org; Marek et al., 2011.

How will we test one versus two progression dimensions?

Model 1 · One-stage model

All repeated measurements are represented by **one common latent PD progression stage.**

Model 2 · Two-stage model

Repeated measurements may reflect **two correlated progression dimensions.**

Dopaminergic progression dimension

Anchored by longitudinal **left/right caudate and putamen DaTScan specific-binding ratios**

Structural progression dimension

Anchored or validated by longitudinal **T1-derived grey-matter measures**, provided structural MRI quality checks are satisfied.

Cognitive measures can relate to either dimension or both. The cognitive tests will not be assigned to a biological mechanism in advance.

Iddi et al., 2018; Wen et al., 2026

Six-month analysis plan

Months 1–2 · Prepare the longitudinal dataset

- Finalize participant inclusion/exclusion criteria
- Reproduce the published CSS benchmark
- Complete structural MRI quality and missing-data checks
- Characterize longitudinal measurement trajectories

Months 3–4 · Fit and compare the models

- Fit the one-stage and two-stage models
- Compare prediction on unseen participant data
- Estimate how cognitive measures relate to the progression dimensions
- Complete prespecified sensitivity analyses

Month 5 · Clinical and benchmark comparison

- Compare with CSS and simpler imaging summaries
- Test prediction of later **motor** and **cognitive-functional** outcomes

Month 6 · Robustness and dissemination

- Complete robustness analyses
- Prepare figures and reproducible code
- Complete the Neuromatch micropublication